DocSense: A Lightweight OCR Free Vision-Language Model For Structured Document Extraction With Retrieval Augmented Question Answering

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***Abstract*—Most document processing pipelines today involve a OCR first step followed by downstream text analysis, which discards layout information with multi-column reading order and table structure, and any errors in OCR propagate to later stages. OCR-free Vision-Language Models (VLMs) such as Donut and Nougat process the page image directly but have limited capability with mathematical content and charts. DocSense is our initial exploration of this architecture on modest hardware, where we finetune a pretrained document VLM, NuExtract3 to produce markdown and LaTeX and add a Retrieval Augmented Generation (RAG) layer to support natural language querying of the extracted content. The paper describes the architecture, the data pipeline used, the 4-bit QLoRA fine-tuning process and the evaluation methodology based on BLEU/ROUGE, edit distance and a custom Visual QA set against an OCR+LLM baseline. It also documents the engineering process (requirements, UML, cost estimation and risk management) and preliminary evaluations. An end-to-end evaluation follows on the full pipeline.**

***Index Terms*—Vision-Language Models, OCR-free document understanding, Retrieval-Augmented Generation, structured information extraction, document intelligence.**

I. Introduction

Research papers, invoices and technical reports are typically dense mixtures of text, spanning multiple columns, tables and mathematical content. The canonical approach to such documents is to run OCR or a PDF parser to extract text for further downstream processing. However, this approach discards layout information that determines multi-column reading order or table structure and any errors in the OCR process are propagated to downstream steps [1].

An OCR-free approach avoids this issue by operating directly on the page image. Donut [1] is an early example of such a system, which trains an encoder-decoder Transformer to predict structured text from page images, with no OCR step in between. This approach was extended to academic documents with LaTeX math in Nougat [2], which acknowledges in its evaluation that the model has limited capacity with mathematical content and encounters ocasional issues with repetition-collapsing. Docsense is our initial foray into this class of models on a single academic GPU, with three main contributions. First, instead of training a Vision Transformer and decoder from scratch we fine-tune an existing document VLM (NuExtract3) to produce markdown and LaTeX content. Second, we attach a Retrieval Augmented Generation (RAG) layer to the document extraction pipeline to support natural language querying of the extracted content. Third, we document the engineering process including the requirements, UML, cost estimation and risk management for the project, in addition to preliminary evaluations.

A. Contributions

Our main contributions can be viewed from three perspectives. First, we propose an OCR-free document processing pipeline that leverages a vision language model, fine-tuned for document understanding rather than trained from scratch. Second, we demonstrate how such a pipeline can be attached to a production chat system with minimal modifications (described in detail in Sec. III-C). Third, we design an evaluation methodology that focuses on the failure modes identified in the literature for similar approaches (Sec. V).

II. Related Work

Donut [1] is one of the earliest end-to-end OCR-free Transformers for document understanding. It learns to predict structured text from scanned document images with a text generation objective. The evaluation focuses on business documents such as receipt, forms and business cards, with less emphasis on scientific documents with multi-column layouts and mathematical content.

Nougat [2] builds on this work by learning to preserve mathematical content in the converted LaTeX representations of PDF pages. The evaluation highlights the model's limitations on mathematical content (higher edit distance) and its occasional failure with repetition-collapsing, which is mitigated (but not eliminated) during training with perturbations. Pix2Struct [3] is another OCR-free approach to document understanding that pretrains a vision-language model on the task of parsing masked screenshots into their HTML counterparts. This demonstrated that the approach was viable beyond document understanding tasks.

BLIP-2 [4] is notable for a lightweight Querying Transformer (Q-Former) that connects a frozen vision encoder to

a frozen large language model. This is similar in spirit to the adapter layer we describe in.DocSense before deciding to pursue a fine-tuning based approach (Section IV). None of these approaches directly address the problem of document extraction with retrieval-augmented question answering which is the focus of this paper.

III. System Architecture

DocSense consists of two stages, depicted in Fig. 1 and described in more detail below.

A. Extraction Pipeline

A page from a PDF is rendered as a 300 DPI image using PyMuPDF and passed to NuExtract3, a pretrained vision language model for OCR-free document extraction to markdown and LaTeX. The model predicts text and tables in Markdown and LaTeX math where applicable.

B. Retrieval-Augmented Chat Layer

The extracted Markdown is split into overlapping 1000 character chunks with 200 character overlap and encoded with Ollama's nomic-embed-text model. If unavailable, a local HuggingFace ONNX sentence-transformer is used as a fallback option. The chunks are stored in a ChromaDB vector store with a local JSON cosine similarity DB as a fallback option. At query time, the question is similarly embedded and the top-k matching chunks are retrieved and used to generate an answer with a hosted LLM (Groq/Llama-3.3-70B).

C. Integration Point

The two stages are connected at a single point. In the chat application, the initial PDF parsing step is substituted with a call to the extraction pipeline. Any files that are already in Markdown are unaffected. This allows individual components to be evaluated in isolation.

IV. Methodology and Implementation

The architectural decisions were not made in stone. An initial design was heavily inspired by Donut [1] and involved training a Vision Transformer, custom adapter layer and decoder from scratch. This was replaced with fine-tuning a document-specialised VLM (NuExtract3) that required less resources.

Fine-tuning was done with 4-bit quantized Low Rank Adaptation (QLoRA) to reduce memory consumption and enable training on a single GPU (NVIDIA RTX 4050 with 6 GB VRAM). We kept the number of trainable parameters low for the same reason. The changes were documented as a requirements addendum rather than updating the Software Requirements Specification (SRS) because the new design fulfilled the same requirements.

The training data consisted of rendered PDF pages from a hand-curated set of categories: text, multi-column, tables and equations. In cases where a category had insufficient representation, subsets of standard document understanding benchmarks were used.

V. Evaluation Methodology

We evaluated the system based on the failure modes identified in the literature. We report BLEU and ROUGE scores for text extraction (compared to hand-curated ground truth) and Levenshtein edit-distance for table and LaTeX structure in each category. Math structure predictions were found to have significantly higher edit-distance than text [2] and were reported separately. A custom Visual QA set was used to evaluate the chat layer for numeric and chart-based questions, which are not directly supported by either Donut or Nougat. For repetition-collapsing, an n-gram repetition detector was run over the model outputs and the percentage of affected pages was reported, compared to the rate for Nougat. Finally, everything was compared to a baseline of Tesseract OCR followed by a text-only LLM on the same test set.

A note on reporting: the fine-tuning process and end-to-end evaluation were ongoing as this paper was being written. We therefore report preliminary results and findings rather than final results. In particular, Section VI describes preliminary validation and engineering analysis, rather than final results. This allows us to publish early findings without overstating the results.

VI. Preliminary Validation and Engineering Analysis

Before final testing and validation, some preliminary analysis was performed on individual components. The NuExtract3 model was validated on the target hardware with 4-bit quantization enabled. Rendering was benchmarked at various DPI settings and 300 DPI was selected as the minimum viable resolution (based on text and table border legibility) with less than 0.4 seconds rendering time per page. Storage size was not found to be an issue. We also validated the integration point by confirming that Markdown input followed the same file parsing code as plain text.

For cost estimation, Function Point Analysis was performed on the data flow diagram to arrive at 72 Unadjusted Function Points with an Adjusted FP of 73.44. An estimated size of 4 KLOC was cross verified (within 1%) using a standard backfire conversion. Using basic COCOMO (organic mode) this corresponded to 10.29 person-months while Intermediate COCOMO with cost drivers for product complexity and less experienced team with fine-tuning (rather than training) accounted for 12.66 person-months.

A Detailed COCOMO analysis at the module level assigned the majority of the effort to the fine-tuning process rather than the overall system. Risk Exposure analysis (probability x cost) arrived at similar conclusions, with GPU memory limitations and model fine-tuning being the main risks.

VII. Limitations and Future Work

DocSense inherits the limitations of NuExtract3, which are likely to impact performance on mathematical content the most [2]. It also has limited support for charts, graphs and plots. Future work should include addressing these limitations and extending the system to larger document sets beyond single

pages. A natural extension is to apply the same method to a multilingual document set from the same source domain.

VIII. Conclusion

DocSense demonstrates that a document extraction system can be built at an academic scale by fine-tuning a document-specialised VLM rather than training a model from scratch and that it can be integrated to a retrieval-augmented chat system with minimal modifications. Evaluation is focused on the failure modes of prior work (repetition-collapsing in Donut/Nougat) and therefore has direct comparability with the literature.

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